CSA0412-OPERATING SYSTEMS

EXP 08-Implement a CPU scheduling program using shortest job first

```
#include <stdio.h>

int main() {
    int n, i, j, temp;
    int bt[20], wt[20], tat[20], p[20];
    float avgwt = 0, avgtat = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    for (i = 0; i < n; i++) {
        p[i] = i + 1;
        printf("Enter burst time for P%d: ", p[i]);
        scanf("%d", &bt[i]);
    }

    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;

                temp = p[i];
                p[i] = p[j];
                p[j] = temp;
```

```
            }
        }
    }

    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
        avgwt += wt[i];
        avgtat += tat[i];
    }

    printf("\nProcess\tBurst Time\tWaiting Time\tTurnaround Time\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t\t%d\t\t%d\n",
               p[i], bt[i], wt[i], tat[i]);
    }

    printf("\nAverage Waiting Time = %.2f\n", avgwt / n);
    printf("Average Turnaround Time = %.2f\n", avgtat / n);

    return 0;
}
```

```
Enter number of processes: 3
Enter burst time for P1: 5
Enter burst time for P2: 3
Enter burst time for P3: 2

Process Burst Time      Waiting Time    Turnaround Time
P3      2               0               2
P2      3               2               5
P1      5               5               10

Average Waiting Time = 2.33
Average Turnaround Time = 5.67


...Program finished with exit code 0
Press ENTER to exit console.
```